

## Assignment: Linux

### Module: 1 - Linux server - Understand and use essential tools

**1) What is the minimum number of partitions you need to install Linux?**

**Answer:**

The minimum number of partitions required is one root partition (/).

This partition contains the operating system files and directories.

**2) Explain About Chmod Command**

**Answer:**

chmod is a Linux command used to change file and directory permissions.

It controls read (r), write (w), and execute (x) permissions.

**Example:** `chmod 755 file.txt`

**3) How to check Linux memory utilization**

**Answer:**

Use the following command:

- `free -h`

It displays total, used, and free memory in human-readable format.

#### **4) Use grep to search for specific patterns in files**

**Answer:**

The grep command searches for text patterns inside files.

**Example:**

- `grep "hello" file.txt`

#### **5) Get Connecting on a Linux server by SSH**

**Answer:**

Use the ssh command to connect remotely to a Linux server.

**Example:**

- `ssh username@ip_address`

#### **6) Create 5 files in the /tmp directory, and then use tar and gzip to compress them**

**Answer:**

**Create files:**

- `touch /tmp/file1 /tmp/file2 /tmp/file3 /tmp/file4 /tmp/file5`

**Compress files:**

- `tar -czvf files.tar.gz /tmp/file*`

#### **7) Describe the root account**

**Answer:**

- The root account is the administrator account in Linux. It has full access to all files, commands, and system settings.
- Root users can install software, manage users, and configure the system.
- Improper use of root privileges can affect system security.

## **8) What is Shell?**

**Answer:**

- **A shell is a command-line interface between the user and the Linux operating system.**
- **It accepts user commands and sends them to the kernel for execution.**
- **Examples include Bash, Sh, and Zsh.**
- **Shells are used for system administration and scripting.**

## **9) What is Linux?**

**Answer:**

- **Linux is an open-source operating system based on Unix. It is widely used for servers, networking, cloud computing, and development.**

**Linux is secure, stable, and supports multi-user functionality.**

**Popular distributions include Ubuntu, CentOS, and Fedora.**

## **10) What is Bash?**

**Answer:**

**Bash stands for Bourne Again Shell.**

**It is the default command-line shell in many Linux distributions.**

**Bash is used to execute commands and create shell scripts.**

**It provides automation and system management features.**

## **11) You have a new empty hard drive for Linux. What is the first step?**

**Answer:**

**The first step is to create disk partitions using tools like fdisk or parted.**

**Partitions organize storage space for Linux installation and file systems.**

**12) Write the Linux command to show the current working directory**

**Answer:**

**pwd**

**13) Write the Linux command to get help with various options**

**Answer:**

**man command\_name**

**Example:**

- **man ls**

**14) Write the Linux command to display what all users are currently doing**

**Answer:**

- **w**

**15) Write the Linux command to get information about the operating system**

**Answer:**

- **uname -a**

**16) Write the Linux command to create a hard link of a file**

**Answer:**

**ln originalfile hardlink**

**Example:**

- **ln file1.txt file2.txt**

**17) Write the Linux command to create a soft link of a file as well as directory**

**Answer:**

**ln -s targetfile softlink**

**Example for file:**

- **ln -s file1.txt linkfile**

**Example for directory:**

- **ln -s /home/user/docs mydocs**

**18) Write the Linux command to search for specific pattern in a file**

**Answer:**

**grep "pattern" filename**

**Example:**

- **grep "Linux" notes.txt**

**19) Write the Linux command to show the use of basic regular expressions using grep command**

**Answer:**

**Example command:**

- **grep "^a" file.txt**

## **Module :2- Linux server - Operate running systems**

### **20) View running processes with ps**

**Answer:**

The ps command is used to display running processes in Linux.

**Example:**

**ps -ef**

- -e shows all running processes.
- -f displays detailed information like PID and user.

### **21) Terminate processes with kill**

**Answer:**

The kill command is used to stop or terminate a running process using its Process ID (PID).

**Example:**

- kill 1234

--1234 is the PID of the process.

### **22) Use top or htop to monitor system resources and processes**

**Answer:**

top and htop are Linux monitoring tools used to view CPU, memory, and running processes in real time.

**Using top:**

- top

**Using htop:**

- htop

**23) Configure one of your lab computers to boot to the CLI using systemd, and reboot to confirm success**

**Answer:**

**Step 1: Set system to boot into CLI mode**

- **sudo systemctl set-default multi-user.target**

**Step 2: Reboot the system**

- **sudo reboot**

**Step 3: Verify CLI mode after reboot**

**After restarting, the system should boot directly into the command-line interface instead of the graphical desktop environment.**

**To switch back to GUI mode:**

- **sudo systemctl set-default graphical.target**

## **Module :3- Linux server - Configure local storage Assignment**

**24) Learn about different filesystem types (e.g., ext4, NTFS)**

**Answer:**

**1. ext4 (Fourth Extended File System)**

- **Default Linux filesystem.**
- **Fast, stable, and supports large files and partitions.**

**2. NTFS (New Technology File System)**

- **Default filesystem for Windows.**
- **Supports file permissions, encryption, and large storage.**

**3. XFS**

- **High-performance filesystem used in Linux servers.**
- **Best for large-scale storage systems.**

**4. FAT32**

- **Compatible with Windows, Linux, and USB drives.**
- **Supports smaller file sizes compared to NTFS.**

**25) Manage disk partitions and filesystems using tools like fdisk, mkfs, and mount**

**Answer:**

**Create partition using fdisk:**

- **`sudo fdisk /dev/sdb`**

**Create filesystem using mkfs:**

- **`sudo mkfs.ext4 /dev/sdb1`**

**Mount partition:**

- **sudo mount /dev/sdb1 /mnt**

**26) Create a 2048MB partition and verify if the partition has been created**

**Answer:**

**Step 1: Open disk using fdisk**

- **sudo fdisk /dev/sdb**

**Step 2: Create new partition of 2048MB**

- **Press n for new partition**
- **Enter partition size:**  
**+2048M**

**Step 3: Save changes**

- **w**

**Step 4: Verify partition**

- **lsblk**

**27) Why LVM is required?**

**Answer:**

- **LVM (Logical Volume Manager) is used for flexible disk management in Linux.**
- **It allows resizing partitions without restarting the system. LVM supports disk snapshots, storage pooling, and easy management of large storage.**
- **It improves scalability and storage administration.**

**28) How can you find out how much memory Linux is using?**

**Answer:**

- **free -h**

**This command displays total, used, and free memory in human-readable format.**

**You can also use:**

- **top**

**or**

- **htop**

**for live monitoring.**

**29) What is a typical size for a swap partition under a Linux system?**

**Answer:**

**A typical swap partition size depends on system RAM.**

- **Systems with 2 GB RAM → 2 GB swap**
- **Systems with 4 GB or more RAM → 2 to 4 GB swap**
- **For hibernation, swap size should be equal to RAM size.**

**Swap is used as virtual memory when RAM becomes full.**

**30) What is the maximum file size on the ext4 file system?**

**Answer:**

- **The maximum file size supported by the ext4 filesystem is 16 TB. It also supports volumes up to 1 EB (Exabyte) depending on block size.**

**31) What is the maximum file size on the xfs file system?**

**Answer:**

- **The XFS filesystem supports a maximum file size of 8 EB (Exabytes).  
It is designed for high-performance systems and large storage environments.**